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BRINGING INCOME DISTRIBUTION IN FROM THE COLD*

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I. INTRODUCTION: THE SUBJECT OF INCOME DISTRIBUTION

The title of this Presidential Address is chosen to highlight the way in which the subject of income distribution has in the past been marginalised. For much of this century, it has been very much out in the cold. There are signs that in the 1990s it is being welcomed back, and I shall be referring to recent research, but I would like to use this occasion to give further impetus to the re-incorporation of income distribution into the main body of economic analysis.

The peripheral nature of income distribution has long been a concern. In 1920, Hugh Dalton wrote in the Preface to his book *Some Aspects of The Inequality of Incomes in Modern Communities* that:

‘While studying economics at Cambridge in 1909–10, I became specially interested in those [parts] which set out to discuss the distribution of income. I gradually noticed, however, that most “theories of distribution” were almost wholly concerned with distribution as between “factors of production”. Distribution as between persons, a problem of more direct and obvious interest, was either left out of the textbooks altogether, or treated so briefly, as to suggest that it raised no question, which could not be answered either by generalisations about the factors of production, or by plodding statistical investigations, which professors of economic theory were content to leave to lesser men.’ (1920), p. vii).

Of course, income distribution was a subject of central importance to classical economists. There is the famous quotation from Ricardo in which he told Malthus that Political Economy should be

* Presidential Address to the Royal Economic Society, Swansea April 1996. This Address is dedicated to the memory of Professor James Meade, who sadly died on 22 December 1995. From him I first learned how economic analysis can help us understand the distribution of income and can contribute to raising the seriousness of public debate.

I should like to thank the many people with whom I have worked on this subject, and on whose research, including our joint writings, I have drawn heavily. In particular, I owe especial thanks to (in alphabetical order) François Bourguignon, Andrea Brandolini, Frank Cowell, Alan Harrison, John Hills, Stephen Jenkins, Mervyn King, John Micklewright, Brian Nolan, Lee Rainwater, Amartya Sen, Tim Smeeding, Nick Stern, Joe Stiglitz, Holly Sutherland, and Chris Trinder. I am most grateful to the following for their helpful comments on an earlier version of this text: Philippe Aghion, Kenneth Arrow, Patrick Bolton, Andrea Brandolini, Stephen Jenkins, Holly Sutherland, and Steven Webb.

‘an enquiry into the laws which determine the division of the produce of industry amongst the classes who concur in its formation’ (1951 edition, p. 278).

This was the functional, or factor, distribution of income, and much of what can be found today in textbooks under the heading of the ‘Theory of Distribution’ is concerned with the determinants of payments to factors (labour, land and capital). In mainstream economic theory, the competitive theory of factor pricing determines the division of national income between wages, profit and rent. Competitive theory has been criticised, with alternatives proposed, such as the Cambridge theory based on the accumulation relationships, or the Kaleckian theory based on imperfect competition, but it is these ideas which form the main component of the theory of distribution. However, as Dalton observed, the relationship of the factor distribution with the personal distribution of income is typically not spelled out. Statements about the division of national income between wages and profits do not tell us directly what determines the share of the top 20% or the bottom 20% of income recipients. The factor distribution is certainly part of the story, but it is only part, and the other links in the chain need to receive attention.

Nor has the personal distribution of income been a central subject for research in the economics profession. An analysis of the contents of this JOURNAL over the past 50 years indicates that, on average, the JOURNAL published one and a half articles a year on income distribution, out of an average of 38 articles per year. In other words, about 4% of the articles dealt with income distribution (broadly interpreted), as shown by the nine-year moving average in Fig. 1. As a basis for comparison, I took international

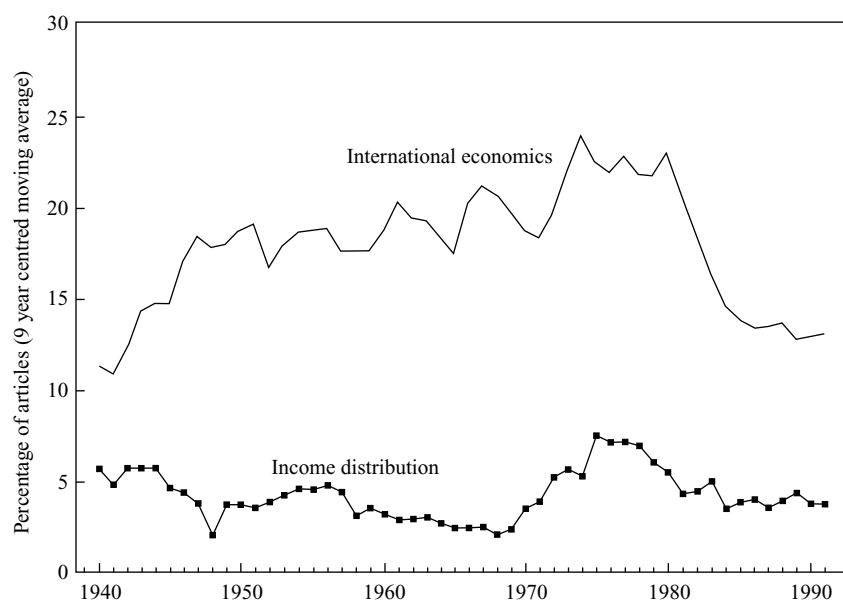


Fig. 1. Articles in this JOURNAL on income distribution and international economics.

economics, which is another significant field, of central importance for the United Kingdom. Here the corresponding average is six and a half articles per year – or about four times as many.

Taking all journals and books together, there is, of course, a large number of articles on income distribution. If one types in the key words ‘income distribution’ to the EconLit database 1969–6/1995, then one comes up with 4,549 entries. (In contrast, ‘international trade’ generates twice as many entries.) But if one examines these, one discovers that a large proportion deal with development economics. This is clearly of great importance, but here I am concerned with OECD countries. A sizeable number deal with the impact of income distribution on other variables. There are articles on the statistical evidence about distribution and about the measurement of inequality; there are articles on the redistribution of income and social security. But what I missed when I read through these entries is research which ties income distribution centrally into analysis as to how the economy works. What is the connection between income inequality and the macro-economic variables that are centre stage in most economic debate? What is the inter-relationship between economic performance and income distribution? How can we use economic theory to explain what is happening to the incomes of individuals, families and households?

II. EMPIRICAL POINTS OF DEPARTURE

In setting the scene for an analysis of the economics of personal income distribution, I begin with empirical evidence about the distribution of disposable household income in the United Kingdom and other OECD countries.¹ I should stress that I am not here attempting to set out the strengths and weaknesses of the evidence on income inequality. There are many limitations to the data presented. They tell us nothing about expenditure, only about income; they omit important sources of income such as fringe benefits or capital gains or undisclosed earnings from the informal economy; they omit the benefits of government spending other than cash or near-cash transfers; they relate to the household and do not explore what happens within the family. When I compare changes over time in income inequality in different countries, the figures are drawn from national studies of income inequality which are not designed for purposes of international comparison. They are not necessarily based on the same concepts of income or method of calculation or period of time, although I have chosen those series which give a reasonable span of years and which are themselves intended to be consistent over time. (I have also in some cases linked series; the sources are listed in the Appendix.)

The data, nonetheless, tell an interesting story. In particular:

- (i) the United Kingdom stands out for the sharpness of the rise in recorded income inequality in the 1980s;

¹ For fuller information about recent trends in income distribution in OECD countries, see Gardiner (1993), Atkinson (1996a), Atkinson *et al.* (1995), and Hills (1996). On the United Kingdom, see Coulter *et al.* (1994), Goodman and Webb (1994), and Jenkins (1995).

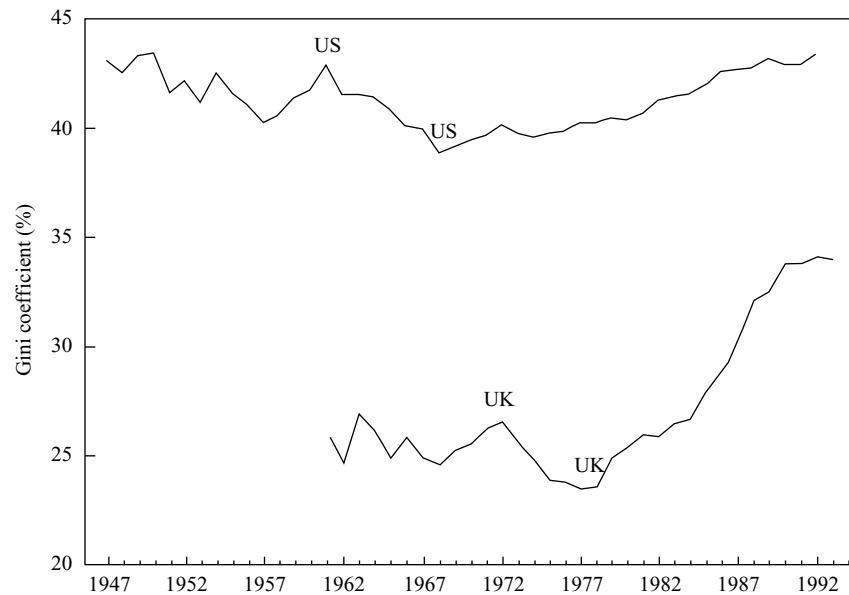


Fig. 2. Income inequality in the United Kingdom and the United States. US 1947–92; UK 1961–93.

- (ii) changes in the personal distribution are large enough to affect our view of aggregate economic performance;
- (iii) changes in inequality may be better described as ‘episodic’ rather than as long-run ‘trends’.

Whether one finds the rise in inequality a matter for concern is a matter of personal judgement. In this paper, I follow conventional practice and refer to income differences as ‘income inequality’, but whether any difference is actually considered an injustice is a matter both of judgement and of interpretation. I am, for example, largely concentrating on snapshots of the distribution – such as income in 1993 or what people earned in the month of April 1995 – whereas in assessing equity we may be concerned with income mobility. We may want to adopt a lifetime or even dynastic perspective, leading us to view the distribution either more or less favourably.

II.A. *Unparalleled Rise in United Kingdom Income Inequality in the 1980s*

One of the most durable of stylised facts in the field of income distribution is the celebrated ‘Kuznets curve’, relating income inequality to the process of industrialisation. Kuznets (1955) considered a two sector economy in which overall inequality depends on the proportion employed in each sector, on the degree of inequality within sectors, and on the difference between the mean incomes in the two sectors. A rise in the proportion employed in the higher income industrial sector could, on certain assumptions, lead first to rising and then falling overall inequality. Kuznets saw the turning point in his tentatively suggested ‘long secular swing in income inequality’ as coming in the last

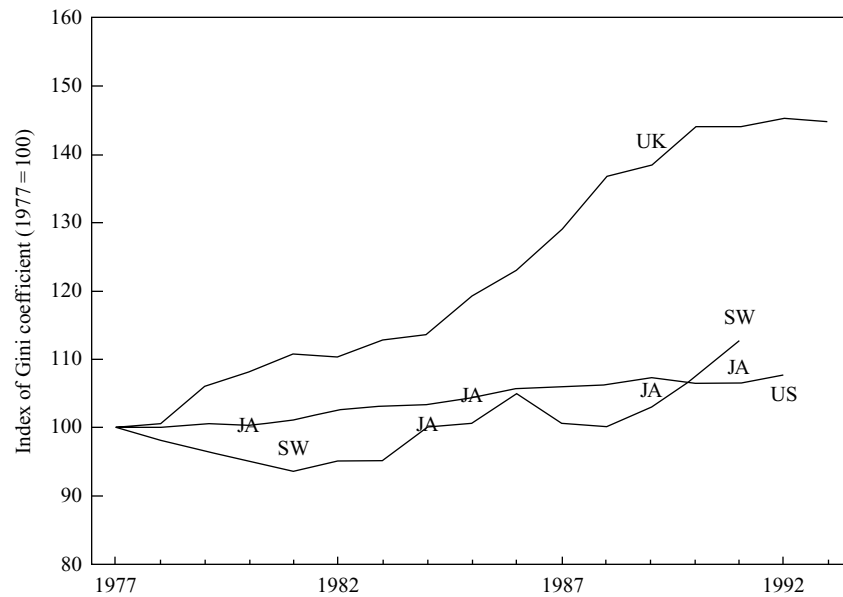


Fig. 3. Income inequality 1977 = 100. UK, US, Sweden (SW) and Japan (JA) (1980 = 100).

quarter of the nineteenth century in England, and somewhat later in the United States and Germany. He was writing in the 1950s, but there continued to be a widespread belief that income inequality in countries such as the United Kingdom and the United States would continue to fall, steadily, if not spectacularly (see, for example, Tinbergen, 1975).

As we now know, income inequality did not continue to fall. In the United States, the Gini coefficient of inequality for household incomes rose between 1968 and 1992 by three and a half percentage points, which more or less took the coefficient back to the level before the decline in the 1960s – see Fig. 2. This is a significant increase, but if you want to see a *big* increase then it is to the United Kingdom that one has to look. Between 1977 and 1991, the United Kingdom Gini coefficient rose by 10 percentage points (these are the estimates of Goodman and Webb, 1994).

Inequality has of course risen in a number of other countries. Fig. 3 shows the change in recorded income inequality since 1977 for Sweden and Japan, in addition to the United States and the United Kingdom. Inequality rose sharply in Sweden at the end of the 1980s, but for a much shorter period than in the United Kingdom. It increased in Japan (shown by isolated points marked JA) but not to the same degree. Nor did inequality rise in all countries over this period. Fig. 4 shows the position for the three large members of the European Union. One clear conclusion is that the United Kingdom stands out for the sharpness of the rise in recorded income inequality in the 1980s. This was unparalleled in the countries examined.

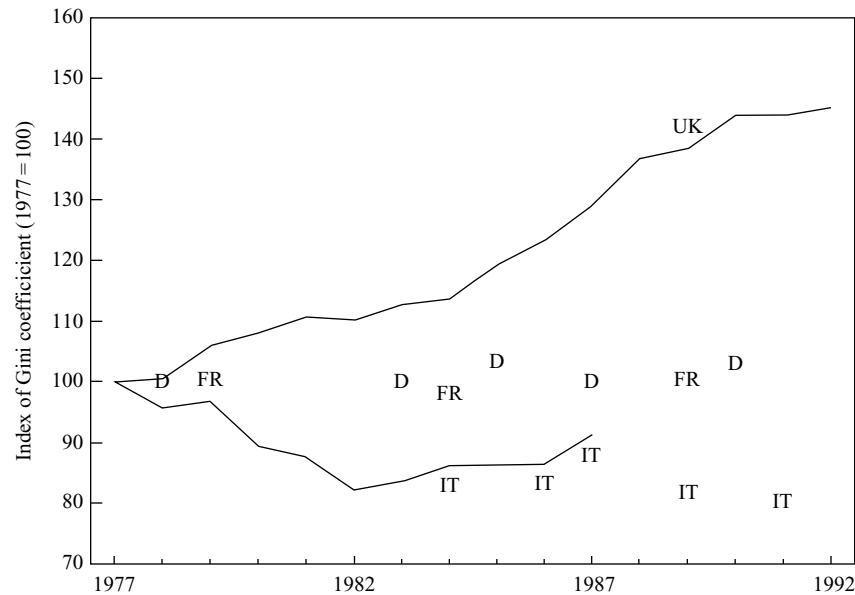


Fig. 4. Income inequality 1977 = 100. UK, (FR) France (1979 = 100), (D) West Germany (1978 = 100) and (IT) Italy.

II.B. *Rising Inequality Matters*

In the past, changes in income distribution have often been dismissed as too insignificant to be worth attention. But this can no longer be done. Changes in the distribution of the magnitude observed in the United Kingdom in the 1980s can affect our view of aggregate economic performance.

The impact of rising inequality does of course depend on judgements of value. A person who is indifferent to changes in the distribution of income is content to measure performance by mean income. The solid line in Fig. 5 shows the trend in mean household equivalent disposable income since 1961, using the estimates of Goodman and Webb (1994). On the other hand, for a person concerned about income inequality, the assessment of the outcome depends on their distributional preferences. The dashed line in Fig. 5 shows the distributionally adjusted real income measure proposed by Sen (1976) on the basis of rank order weights (mean income times (one minus the Gini coefficient)). The distributionally adjusted measure tracks the mean income closely until 1973; it then does a little better until 1979; and after that it falls progressively behind. The Gini adjustment is only one of many possible, but it shows that taking account of distributional changes can give a different picture of the growth performance: the record of the 1980s looks less impressive.²

² Distributional corrections to the UK growth rate have been made by Beckerman (1980) and by Crafts (1993) in his evaluation of the 'Thatcher Experiment'. In the United States, Klasen has shown how distributionally-weighted growth rates 'shed a much more favorable light on improvements in well-being during the 1960s, particularly compared to the 1980s' (1994, p. 270).

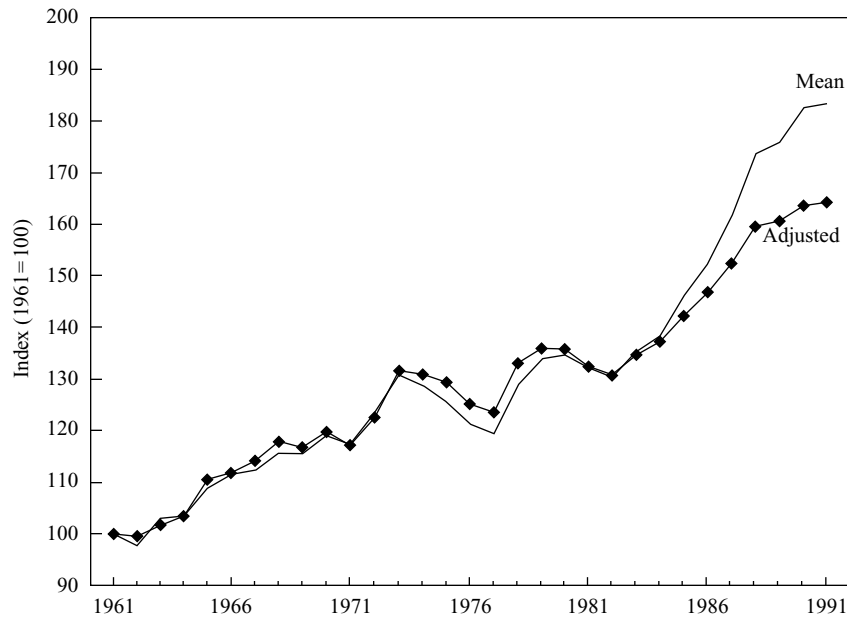


Fig. 5. UK mean income and distributionally adjusted income using Gini coefficient.

II.C. *Episodes not Trends*

Not only has the 'Kuznets curve' been confounded by recent events, but it has also become clear that it is misleading to talk of 'trends' when describing the postwar evolution of the income distribution. The evidence in Figs. 2–4 suggests that it may be better for a number of countries to think in terms of 'episodes' when inequality fell or increased. Returning to Fig. 2, we can see that there was an episode of declining inequality in the United States during the Kennedy/Johnson years (1961–8, marked by US–US), followed by a period of increase from 1969. In the United Kingdom, there appears to have been a decline post-1972 (marked by UK–UK), which may, or may not, be associated with incomes policies (see below); and the post-1977 period divides into three sub-periods, with a faster rise in the Gini coefficient after 1984, and possibly a flattening in the 1990s. Here I should note that the most recent figures from the Central Statistical Office (CSO) show 1991 as being the highwater mark of inequality in the United Kingdom.³

What I am suggesting is not new. Those used to working with macro-economic time series will recognise that I am describing a segmented trend model (Perron, 1989), which does indeed seem to provide a reasonable

³ The CSO estimates differ in definition from those of Goodman and Webb (1994) used in Fig. 2, in that they are weighted by households, are not re-weighted for differential non-response or for the under-representation of high incomes, and in annualising their data in some respects.

The CSO estimates show the Gini coefficient for household equivalent disposable income rising from 27% in 1979 to 34% in 1989 (Central Statistical Office, 1994, p. 123). The CSO estimates for later years differ in including the benefit from company cars and from beneficial loans for house purchase from employers; they show a Gini coefficient of 34% for 1992 and 1993/4 and 33% for 1994/5 (same source and Central Statistical Office, 1995, p. 46).

description of the data. My emphasis on the episodic nature of change in the income distribution has parallels elsewhere. In his account of wage differentials, Reder notes that

‘The long-run decline in the skill margin in advanced countries has not occurred slowly and steadily. Instead, the skill margin appears to have remained constant for relatively long periods of time and then to have declined sharply within a very few years’ (1962, p. 408).

The analysis by Donohue and Heckman (1991) of the status of black Americans concludes that

‘the story of black economic progress is not one of uniform secular advance, but rather of episodic change.’ (1991, p. 1604).

More generally, one advantage of rejecting an evolutionary approach in favour of an ‘episodic’ characterisation, is that as described by Giddens, it

‘helps free us from the tendency...to analyse societal development in terms of “stages”, and from the influence of “unfolding models” of change’ (Giddens, 1981, pp. 82–3).

I should emphasise that episodic change in inequality may be in either direction: the recent United Kingdom experience could be reversed. It also has important implications for the possible lines of explanation, to which I turn in Sections IV–VI. It means that we have to consider not just the long-run equilibrium properties of the distribution, but also the explanations of these episodic departures, if possible making these endogenous.⁴

III. THE SOURCES OF INEQUALITY

One could describe the textbook economic approach as starting from the underlying economic forces and working back to how they impinge on individuals and families. In principle, the line of argument leads from the factor distribution to the personal distribution of income. The trouble is that it often does not seem to get there. The link is not made: we are left wondering about the implications for the personal distribution.

I therefore want to start from the other end: with the sources of household income. According to the *Family Expenditure Survey*, which is the origin of the United Kingdom data I have been using, the bulk of household income comes from work (employment and self-employment), but the proportion has been falling: from 83% in 1973 to 75% in 1983 and 73% in 1993.⁵ Recorded household income from capital rose from 1973 to 1993 (from 7% to 11%), but this has come increasingly through the route of annuities and private pension

⁴ Among the factors which may be associated with episodes of distributional change are shifts in demographic structure. These are not discussed here, but see, in a United Kingdom context, Mookherjee and Shorrocks (1982) and Jenkins (1995).

⁵ These figures are from Central Statistical Office (1994), Chart 8.1, page 85. It should be noted that in these figures people away from work without pay for 13 weeks or less are treated as continuing to receive their normal wage or salary. Although the series over time is shown as continuous by the CSO in this graphic, there is a break in the series in 1983 – see Atkinson (1993).

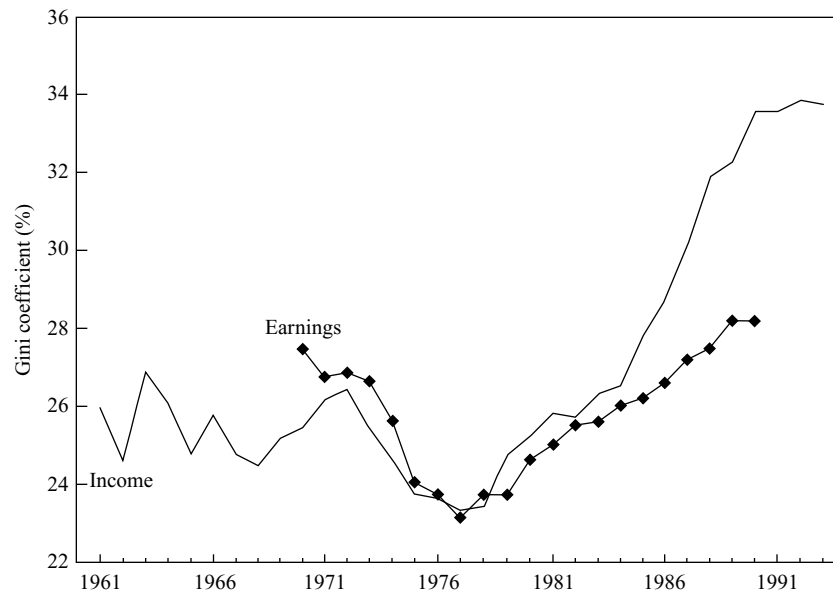


Fig. 6. Incomes and Earnings. Inequality of household incomes and individual earnings.

benefits, which have doubled from 3 to 6 %. Finally, one has to remember that the second largest source of income is social security benefits. This accounted for 14 % in 1993, although the proportion fell between 1983 and 1993, after having increased greatly between 1973 and 1983 (from 9 % to 16 %).

These three sources – earnings, capital income and transfers – will be my focus throughout the rest of the paper.

III.A. *Earnings Dispersion*

When one talks about income inequality, most people think of rising earnings dispersion, and this is indeed the aspect which has received most attention from the economics profession. In the United Kingdom there is plain evidence of widening differentials in the distribution of wage income (see Gosling *et al.* 1996): for all workers, paid for a full week, the real earnings of the bottom decile, deflated by the retail prices index, grew by 11 % between April 1979 and April 1995, compared with 50 % for the top decile.⁶

Fig. 6 shows the movement in earnings dispersion for individual employees and the comparison with the household income inequality series we have been using. The two series appear to move together over the 1970s and early 1980s, but from 1984 to the end of the 1980s there was a divergence, with the income coefficient rising more sharply. The rise in earnings dispersion is a powerful contributing factor but only part of story. Inequality among those in work has to allow for the self-employed, whose importance in the distribution has been stressed by Goodman and Webb (1994), Jenkins (1995) and Parker (1996). But

⁶ These figures are from the *New Earnings Survey* (1979, p. A34 and 1995, p. A1.1), and relate to adult male full-time workers whose pay was not affected by absence. They are adjusted for the change in definition of adult workers in 1983. The price index used is the all items retail prices index.

there was also, particularly between 1975 and 1985, a large rise in the proportion of families without incomes from work: from 20% to 30% (Atkinson, 1993, table 5). As has been stressed by Gregg and Wadsworth (1996), there has been a divergence between the employment rates of individuals and of households, with a rise in both workless and two-income families.

III.B. *Incomes without Work: Capital Income and Transfers*

The rise in the proportion of families without income from work is important because this group not only has a lower average income but also exhibits considerable inequality. It is commonly believed, especially given the largely flat-rate benefit structure, that there is much less inequality within this group, but being without work income does not reduce everyone to the same level: for family units, in 1985 the Gini coefficient was virtually the same as for those with work income (Atkinson, 1993, table 5). This makes it of particular interest to examine the position of this group.

One element is capital income. Real rates of interest increased in the early 1980s and remained high a decade later (see Blanchard, 1993); real dividends increased, and share prices rose still faster. The implications of these changes for the observed distribution of household income are, however, far from transparent: it is money income, rather than real income, which appears in the statistics, and allowance has to be made for financial and other intervening institutions, such as pension funds – see Atkinson (1996*b*). Here I simply note for future reference that the rise in expected real interest rates (Scott, 1993) may well have affected household decisions, an aspect to which I return when discussing investment in human capital.

The main source of income for those not in work is in fact social security, and this brings me to the distributional impact of the government, a subject which goes back at least to Jevons who wrote a memorandum for the Treasury in 1869 (Roseveare, 1973). The present day counterpart is the *Economic Trends* study published by the Central Statistical Office (1995). This shows the difference between ‘private income’ (i.e. earnings plus capital income plus private transfers) and post-tax income allowing for social security transfers received, and direct and indirect taxes paid. We cannot draw any conclusions from this difference as to the actual incidence of the government budget, but it is, nonetheless, interesting that the *Economic Trends* study shows that in the first half of the 1980s the Gini coefficient for private income (not including transfers) increased sharply, with a much more moderate increase in the coefficient for post-tax and benefit income. At first sight, at least, the welfare state appears to have moderated the rise in pre-transfer incomes. After 1984 the situation reverses: the Gini coefficient for original incomes rose by one percentage point from 1984 to 1989 but that for post-tax income increased by seven percentage points.

The apparent fall in the redistributive impact of transfers and direct taxes since the mid-1980s is circumstantial evidence that policy changes have contributed to the rise in income inequality. Redmond and Sutherland (1995)

have calculated that application of the 1978/9 tax and benefit system, indexed in line with per capita GDP, to the 1994/5 distribution of household incomes would have reduced the tax burden for all decile groups except the top; and the Gini coefficient would have been lower by about five percentage points. At a more detailed level, Atkinson and Micklewright (1989) list seventeen distinct changes in unemployment insurance between 1979 and 1988, the majority of which reduced the level or coverage of benefit.

To sum up, it is not just the dispersion of labour income that we need to understand – important though that is – and it is not just private incomes that need to be considered. The determinants of public redistribution are part of what has to be explained. Do they reflect shifts in the constraints faced by governments (for example, on account of fiscal competition)? How far are they the outcome of changes in the prevailing ideology? I return to these questions in Section VI.

IV. DIFFERENT EXPLANATIONS OF EARNINGS DISPERSION

As already indicated, there is at present limited connection between economic theory and the explanation of personal income distribution. This is not to suggest that there is *no* connection, and I begin with one of the areas which has been most discussed in recent years: the explanation of earnings dispersion.

This recent literature is remarkable both for its liveliness and for the extent to which supply and demand considerations hold sway. There appears to be widespread agreement on a straightforward explanation of rising earnings dispersion: there has been a shift in demand away from unskilled labour in favour of skilled workers. In the United States and the United Kingdom this has led to a fall in the relative wage of unskilled workers, and hence a rise in dispersion.

IV.A. *Supply and Demand for Skill*

Increased dispersion is explained therefore by the simplest of economic tools: supply and demand. Suppose that we consider just two kinds of labour (skilled and unskilled). If the premium for skilled workers increases at a time when the relative number of skilled workers in employment rises (as has been the case – see Levy and Murnane, 1992, and Gregg and Machin, 1994), then there must have been a rise in the relative demand for skilled labour. Why should the demand curve have shifted? One much-discussed explanation is the liberalisation of international trade and increased competition from the countries where unskilled labour is abundant: the growth of North–South trade in manufactures, as emphasised by Wood (1994). The precise contribution of trade has been debated (see Burtless, 1995, for a recent review), and other explanations have been advanced, notably that technical change has been biased towards skilled labour with the introduction of automation and Information Technology (IT). Moreover, it is possible that trade and technical change interact.

Whether based on trade or on technology, the supply and demand story

sounds an attractive one, not least because it can be readily explained to a lay audience:

‘factory workers in developed countries face the twin threat of robots filling in job applications, as well as long lines of overseas suppliers offering to produce goods at a fraction of the cost’ (Leamer, 1995, p. 6).

It can also explain why there may be episodes of rising or falling differentials. If, for instance, government education and training policy were to lead to an expansion of the supply of skilled workers, then we would expect the wage premium to fall over a period. If there is major technological innovation, such as the introduction of IT, then this may cause a once-for-all shift in the opposite direction.

The proponents of the skill-based explanation stress that it is not just a matter of observable job characteristics, but also of unobserved skill components. This extension is necessary in view of what Krugman (1994) has called the ‘fractal’ quality of increased dispersion: however narrowly one defines groups, one still finds an increase in dispersion. Evidence for the United Kingdom is provided by Robinson (1994), who reports that the dispersion of male earnings, measured by the decile ratio (ratio of top to bottom decile), widened significantly between 1979 and 1990 for all occupational groups except III (Professional and related in education, welfare and health) and IX (Security and protective service). The more detailed occupational data are shown in Fig. 7, where I have taken the 38 detailed KOS occupations where there were more

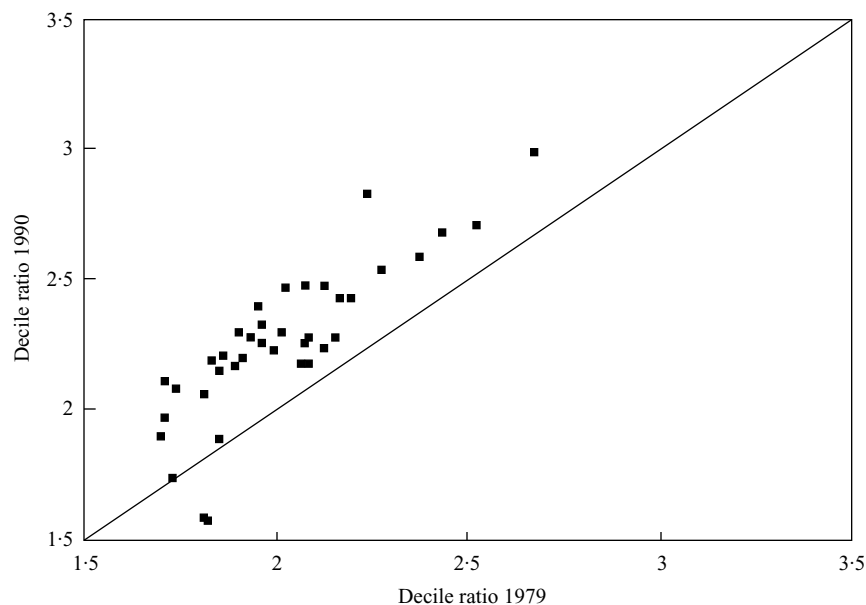


Fig. 7. Occupational groups. Decile ratio of individual earnings within occupational groups.

than 500 employees in 1979 and followed them to 1990, which is the date when the classification was changed.⁷ The figures refer to adult men paid for the full week (no adjustment is made for the change in the definition of 'adult' in 1983). The overall decile ratio was 2.38 in 1979 and 3.11 in 1990. There are two observations below the 45° line: policemen and secondary school teachers. There are two close to the line: further education teachers and bus and coach drivers. But the great majority show increased dispersion even within narrowly defined occupational groups.

Faced with such a fractal picture of differences within narrowly-defined occupational or educational groups, some researchers have concluded that it too is a matter of skill differences: according to Murphy

'the increase in within-group variation is generated by the same forces leading to greater wage inequality across groups (i.e., the general growth in the demand for skill)' (1995, p. 56).

Others may conclude that the increased dispersion within occupational groups is a reason for seeking additional explanations.

IV.B. *Alternative Explanations*

The present-day hegemony of the supply and demand story contrasts markedly with earlier writing on wage differentials, where there has been a creative tension between market force and alternative explanations of wage differentials. Phelps Brown, for instance, opened his *The Inequality of Pay* (1977) by contrasting the 'economist's' approach to pay determination with that of the 'sociologist': the economist sees people as engaged in rational, impersonal transactions; the sociologist sees people interacting as members of a society.

In part, social interactions are institutionalised through collective bargaining and government intervention. The simple competitive model has to be modified to allow for trade union/employer negotiations and for the impact of government, both on collective bargaining and directly on wage determination. If it is the case that unions lead to lower earnings dispersion, then one possible cause of the rise in dispersion in the United Kingdom is the decline in union power and coverage; and there is evidence that part (but only part) can be explained in this way. Gosling and Machin (1993), using establishment data on semi-skilled earnings, conclude that decline in unionisation accounted for around 15–20% of increased earnings dispersion in the 1980s. A similar result is reached by Bell and Pitt (1995) using Family Expenditure Survey data on individual male earnings.

The direct impact of government in the 1980s has included the removal of 'fair wages' requirements for government contractors and the abolition of Wages Council protection for lower-paid workers. Dickens *et al.* (1994), using New Earnings Survey data, find that Wages Councils significantly compressed the distribution of earnings in Wages Council industries. More generally, in the

⁷ The KOS data have been examined by Bell *et al.* (1990). For instance, they show for engineering technicians that over the period 1973–82 there was first a fall then a rise in dispersion which was less pronounced than, but mirrored, the overall change.

1970s Government incomes policies were explicitly concerned to tilt pay settlements in favour of the less well paid (for a summary, see Goodman and Webb, 1994, p. 17). The Conservative Stage Two in 1973 set a group pay limit of £1 plus 4%, with an individual maximum increase of £250 a year. Labour's *Attack on Inflation* in 1975 restricted increases to £6 a week, with no increase for those earning more than £8,500 a year.

Incomes policies are examples of what I mean by 'episodes'. If we were to carry out a time series analysis of the earnings dispersion series, then the imposition and ending of incomes policy would provide natural points at which to introduce a break in the trend. However, the extent to which incomes policies in fact achieved their redistributive aims has been the subject of debate. While some argue that reduced earnings dispersion in the 1970s was, at least in part, the result of incomes policy (see Chater, 1981, for a case study of the engineering industry), and that the subsequent reversal reflects its abandonment (Adams, 1988), others have been more sceptical, pointing to the lack of correspondence between the particular policy stages and the observed narrowing (Dean, 1978) and to the failure to observe the same outcome at the level of individual settlement groups (Ashenfelter and Layard, 1979). Brown (1976 and 1979) has argued that the levelling during the 1970s was more associated with high rates of inflation, combined with concepts of 'fairness' involving absolute rather than relative positions.

IV.C. *Social Custom and Norms*

Trade union bargaining and statutory wage determination may be incorporated into the supply and demand framework, or the economists' approach, and the same applies to monopsonistic behaviour by employers. More of a challenge is the view that supply and demand only place limits on the possible wage differentials, with other factors such as social norms determining where between these limits wages actually lie. Such a 'range theory' of wage differentials was advanced by Lester (1952) and has long been implicit in much institutional writing on labour economics, even if it has received less attention in recent years. Within this range, there is scope for notions of fairness or equity, as has been investigated by, among others, Wood (1978), and Carruth and Oswald (1989).

It seems to me that these alternative approaches have merits which have been too hastily discounted by supply and demand theorists. There are good grounds to try and build bridges. Such an approach has been well illustrated by the research of Akerlof and of Solow on involuntary unemployment. From their work, one can draw the lesson that observance of social norms can be consistent with individual rationality, even where it may appear to conflict with economic advantage. Akerlof (1981) describes a model where individual utility depends not only on income but also on reputation and, for those who believe in the social code, on conformity with the code. The loss of reputation depends on the proportion who believe in the code, which is undermined if people cease to observe it. He shows that there may be a long-run equilibrium with the persistence of a 'fair', rather than market-clearing, wage. Solow

(1990) uses a repeated game model to argue that it may be individually rational for unemployed workers not to undercut the wages of those in employment.

In this way, we are not suspending supply and demand as much as enriching the behaviour which lies behind these relations. Looked at this way, widening wage dispersion can result not just from shifts in the demand for skill but also from changes in social norms. It may, for exogenous reasons, have become socially acceptable to have larger wage differentials within the workplace. Or the exogenous shift in demand may have interacted with the endogenous determination of social norms. As more people are remunerated outside the conventional norms, so adherence to these norms becomes weaker, or the socially acceptable range widens. There may be multiple equilibria. The demand shift may have caused a movement from a low differential equilibrium to a high differential equilibrium. This line of explanation seems to me to repay further consideration, drawing on sociological and psychological theories of social norms.⁸

V. OTHER PARTS OF THE PUZZLE

The account given so far may be criticised as partial/partial analysis. It is partial in the sense that it has focused on the labour market, and not considered the general equilibrium of the economy as a whole. It is partial/partial in that relatively little has been said about the supply side. If differentials widen on account of demand shifts (or changes in social norms), what effect will this ultimately have on the supply of workers with different skills?

The potential importance may be seen from a simple model. Suppose that ability differences affect earnings equally in skilled and unskilled jobs, that there are no other costs of training apart from the time spent acquiring the skill, that everyone can borrow at an interest rate r , and that the working life is the same. Then for the skilled wage, w_s , to compensate exactly for the delayed entry into work, it has to be the case that $w_s e^{-rS} = w_u$ where the length of training is S , and w_u is the unskilled wage. In terms of the supply and demand for skill diagram, the relative supply curve in terms of relative wages is horizontal; in the long-run, where the wage differential (w_s/w_u) is equal to e^{rS} , people are indifferent between skilled and unskilled jobs. In the long-run, shifts in demand affect the number of skilled workers but not the wage differential.

In such a case, the differential exactly compensates for the cost of education (delayed earnings). This has two important implications. First, no lifetime inequality is introduced. This simple observation is often overlooked in the public debate. It is indeed striking how much the recent discussion has focused exclusively on wage differentials and not asked whether such differences are associated with inequality. This re-inforces the warning given earlier that, although concentrating on a snapshot of the distribution, we need to bear in mind the lifetime perspective. Secondly, the compensating wage differential depends on the rate of interest, so that if real interest rates have risen this may

⁸ An alternative approach is to make endogenous people's beliefs about the relation between their actions and economic rewards – see Piketty (1995*b*).

explain part of the observed widening in the wage distribution. This takes us outside the labour market. We are led to ask how people are differentially affected by a rise in the interest rate, which in turn depends on their initial endowments of capital.

V.A. *The Capital Market and General Equilibrium*

One of the important contributions of James Meade to our understanding of income distribution is that he set the acquisition of marketable skills in the wider context of home background and the transmission of advantage from generation to generation, through both human capital and material inheritance. In *Efficiency, Equality and the Ownership of Property* (1964), he described a model of intergenerational transmission, later developed in 'The Inheritance of Inequalities' (1973). Among other elements, educational attainment was assumed to be affected by parental income and wealth, moderated by stochastic elements ('luck') and social contacts. Property was accumulated through saving and inheritance, and the rate of return to savings was assumed to be an increasing function of wealth on the grounds that the fixed costs of acquiring information could be spread. This illustrates the 'positive feedback' emphasised by Meade:

'self-reinforcing influences which help to sustain the good fortune of the fortunate and the bad fortune of the unfortunate' (Meade, 1976, p. 155).

Meade's microeconomic analysis of income distribution was not explicitly related to the macro-economy, but Stiglitz (1969) set the model in the framework of neoclassical growth, where factor returns depend on the stock of capital. With the specific assumptions made (including a proportional savings function and the equal division of estates), Stiglitz proved that, in the absence of intrinsic differences between people, of imperfections in the capital market, and of stochastic elements, the distribution converges to equality. Convergence to long-run equality of wealth is guaranteed by the steady state condition that the rate of return is less than the rate of growth. This result depends on the assumptions. Convergence does not necessarily follow where consumption decisions are based on maximising the infinite stream of dynastic utility (Bliss, 1995). Stiglitz showed that unequal inheritance in the form of primogeniture could lead to sustained inequality. Bourguignon (1981) demonstrated how non-convexity in the accumulation relationship can lead to a two-class equilibrium, with persistent inequality despite people being intrinsically identical.

Non-convexity has been introduced in a different way in a recent interesting series of papers on the macro-economics of income distribution by Aghion and Bolton (1992, 1993), Banerjee and Newman (1991, 1993), Galor and Zeira (1993), and Piketty (1994).⁹ Suppose that we combine the earlier supply and demand model of skill differentials (involving an indivisible investment in training) with a model of imperfections in the capital market, and the transmission of wealth from generation to generation. For this purpose, I

⁹ For a review of these, and other contributions, see Brandolini and Rossi (1995), and Piketty (1995a).

simplify by assuming away stochastic elements, as is done by Galor and Zeira (1993), whose work I am following closely. There are, as in the model of Meade, a sequence of dynasties, although I do not allow for marriage, so that everyone reproduces unaided (and has one child). There are overlapping generations,¹⁰ with bequests made at the end of the second period, so that each dynasty inherits in middle age. Bequests passed on to the next generation are determined as a fraction of lifetime wealth, based on the maximisation of a lifetime utility function where the amount bequeathed enters (a ‘warm-glow’ version of the bequest motive).

People are identical on birth in all respects except for their anticipated bequest; this does, however, affect their decision whether or not to acquire skills. The capital market is imperfect in that people can lend freely at a (continuous) rate of interest r but can only borrow against collateral. In the case of educational finance, expected bequests serve as a collateral. There is a critical level of bequest received below which people cannot afford to finance their consumption during education; moreover, this is an increasing function of the interest rate.^{11,12} It is assumed that we are in a small economy open to world capital and product markets; the interest rate is therefore the world interest rate, but the wages are determined in the labour market where the demand is that of profit-maximising firms with identical production functions (assumed to be Cobb–Douglas).

The outcome depends on the various parameters. Fig. 8 shows a situation like that in the Galor and Zeira analysis where there is a long-run equilibrium with two groups, with different amounts of capital, where the richer group are skilled workers and the poorer are unskilled workers. The initial level of inherited wealth, i , is shown on the horizontal axis in the right hand quadrant. Those with more than i^* have sufficient collateral to invest in education. The lower right hand quadrant shows the distribution in a specified generation of people with wealth below i . If investment in skill is rationed by the capital market constraint, then the proportion below i^* determines the proportion of unskilled workers, denoted by l_u .

The wages for skilled and unskilled labour are shown in the bottom left quadrant as functions of the proportion of unskilled workers. (These functions are derived from the profit-maximising conditions of firms, and depend negatively on the rate of interest.) Comparing w_u with $w_s e^{-rS}$, we can see whether or not people would choose education if unconstrained. The diagram has been drawn in such a way that the constraint is binding, so that the supply

¹⁰ All education and work takes place in the first period, and all work for the same length of time. Skilled workers spend the first fraction, S , of the period being trained, and then work for the remaining $(1-S)$ of the first period. Unskilled workers work for the first fraction $(1-S)$ of the first period, and then retire early. All workers are retired for the second period. The Galor and Zeira model has been adapted in this way to study the impact of pension schemes by Alessandra Casarico of Brasenose College, Oxford, in her M.Phil dissertation.

¹¹ It is assumed that there is a minimum level of consumption which has to be financed by the individual during training, which grows during the training period at exponential rate r .

¹² In Banerjee and Newman (1993) and Aghion and Bolton (1993), people borrow against collateral to invest in entrepreneurship; in both cases the minimum wealth level to make the investment is an increasing function of the rate of interest on a safe asset. See also Ferreira (1995).

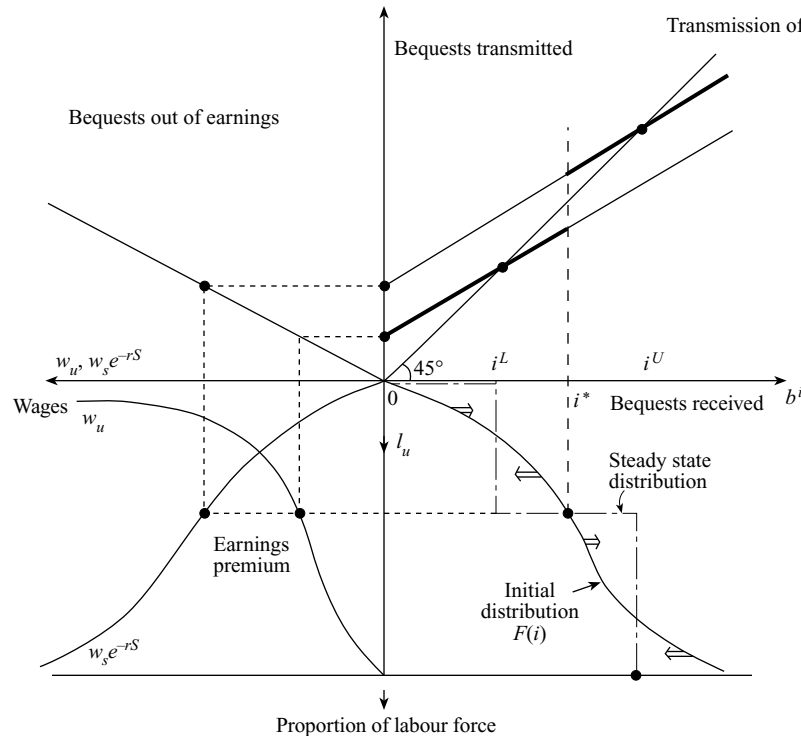


Fig. 8. Distribution of income with imperfect capital market. $\Leftarrow$ shows evolution of distribution across generations.

of skilled labour consists of those who can borrow to finance the acquisition of education, and the wage premium exceeds the compensating differential.

The wage premium gives an advantage to skilled workers in terms of lifetime earnings which feeds into the determination of bequests out of earned income in the top left hand quadrant. In turn, this determines the intercept in the overall bequest relationship, and hence the wealth inherited by the next generation – see the top right hand quadrant in Fig. 8. From this, we can see how the distribution evolves over time. With the combination of parameters shown, the initial class division is maintained, with people initially below i^* converging to i^L and people initially above i^* converging to i^U (shown by the dashed and dotted lines).

My object in this paper has been to incorporate income distribution into the mainstream of economics, and what could be more mainstream than a four-quadrant diagram? Moreover, as in other branches of economics, it yields interesting comparative statics and dynamics. For instance, we can follow through the general equilibrium implications of technical change affecting the relative demand for skilled and unskilled labour, which would shift apart the curves in the bottom left hand quadrant. The model can be used to investigate the consequences of a rise in the real interest rate. It affects the demand for labour, shifting the w_u and w_s curves inward (we are moving round the factor/price frontier). The rise in r increases the compensating wage premium

e^{r^S} . In the top left hand quadrant, the propensity to bequeath rises, as does the slope in the top right hand quadrant. At the same time, the rise in r increases the necessary collateral, and hence i^* .

Or to take a simpler exercise, suppose that we start, not from Fig. 8, but from Fig. 9. Here the cost of education is lower, so that i^* now lies below the value at which the unskilled class are in equilibrium. The unskilled are subject to the capital market constraint, so the wage premium exists, but over generations their wealth is rising, so that eventually the point is reached where the capital market ceases to be a constraint and the wage differential is at the equilibrium level. We are heading towards a situation where there is only one class.

Suppose now that this benign process (benign not least because ultimately the capital market imperfection ceases to be operative) is interrupted by an upward shift in the cost of education (for example, as a result of eliminating state subsidies). If sufficiently large, then this could transform the dynamic evolution, with people in the lower class unable to accumulate sufficient collateral. They would become trapped, as in Fig. 8. The whole nature of the distribution would change.

The model just described falls well short of incorporating all the rich detail of Meade's account of the determination of incomes (there is no marriage in the model, nor differential family size, nor genes, nor social contacts) and it does not do justice to important strands in the recent literature.¹³ It does however, cast light on a number of current issues, including the phenomenon noted in the 1995 OECD survey of the United Kingdom that

'economic inequality in general hampers education and training reform. Income distribution has widened significantly since 1979. High income inequality can act to constrain pupil achievements in the lower tail of the distribution: learning can be a struggle for pupils from households which lack the resources to support their learning.' (OECD, 1995, pp. 81–2).

VI. PUBLIC CHOICE

The importance of state transfers in the distribution of personal income means that, as I have argued elsewhere (Atkinson, 1993), we need to go beyond purely economic explanations and to look for an explanation in the theory of public choice, or 'political economy'. We have to study the behaviour of the government, or its agencies, in determining the level and coverage of state benefits. The government's actions cannot be treated as purely exogenous.

There has been a recent resurgence of interest amongst economists in the politics of income redistribution, stemming particularly from concern with the relationship between income inequality and the rate of growth, including Alesina and Rodrik (1991), Bertola (1993), Perotti (1992, 1993), Persson and Tabellini (1994), and Saint-Paul and Verdier (1992). The models differ in their treatment of the link between distribution and growth, but they share a

¹³ For instance, Brandolini (1992) has cast the relation between factor and personal distributions in terms of 'entitlement rules', which determine individual claims on the income from production. A second example of important work not referred to here is that on neighbourhood effects and human capital formation – see for example Durlauf (1996) and Bénabou (1996).

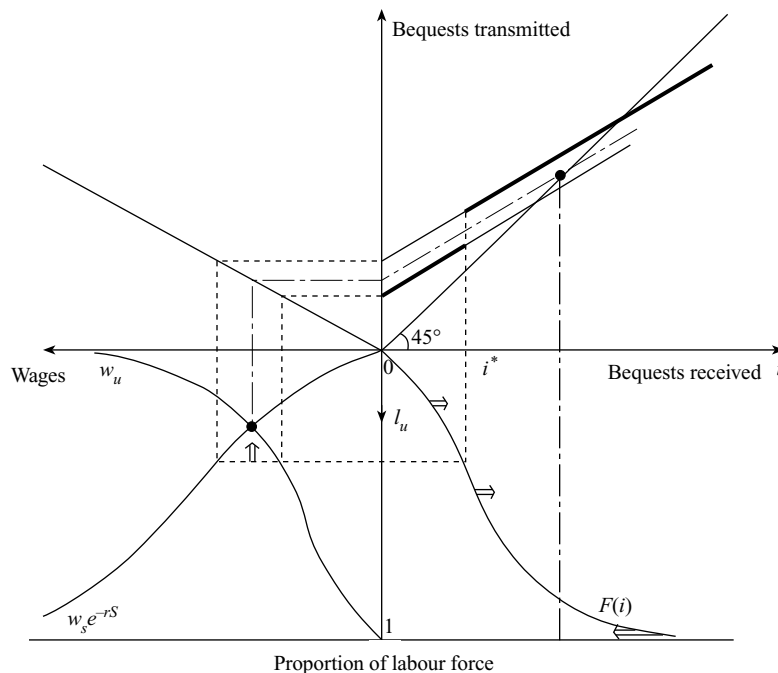


Fig. 9. Evolution towards one class society. $\Leftarrow$ shows evolution of distribution across generations.

common approach to the determination of political equilibrium. As explained by Perotti in his review of this literature,

‘one only needs to specify the political mechanism. In all cases, this is essentially some version of the standard median-voter result.’ (1992, p. 312).

In my view, this understates what economists can usefully learn from political scientists (who may be a little offended by the word ‘only’). The median voter theory is far from being ‘standard’. It seems to me important to see how far the findings depend on whether the outcome is governed by the preferences of the median voter, or by the ideology or preferences of political parties, or by political pressure from different interest groups, or by bureaucratic control of civil servants or agencies. There has been relatively little research by economists which has set side by side different possible explanations of income redistribution, and examined the sensitivity of the conclusions to the choice of model.

Even if we accept that the preferences of voters are decisive, the modelling is a matter of some subtlety, as may be illustrated if we consider a concrete example and ask why has the response to higher unemployment in Britain been to reduce the relative level, and the coverage, of unemployment benefit? Then the median voter model can be interpreted in at least two different ways.

Suppose first that policy towards unemployment benefit reflects the wishes of the majority of the electorate who are in regular employment and who are assumed to have some degree of concern for the level of welfare of the

unemployed. They maximise a function of their own net income and of the level of (flat-rate) unemployment benefit. This may be seen as a welfare function based on solidarity: people in employment are willing to pay a contribution towards unemployment benefit. We may now ask what happens if a labour market shock causes the cost of this policy to rise: for example where there is a rise in the number of beneficiaries, reducing the benefit per head for a given tax rate. This may well lead to a fall in the replacement rate preferred by the majority in work. On the other hand, it is easy to construct examples where the voter would choose the same replacement rate: there is an equi-proportionate reduction in benefits and net earnings (Atkinson, 1990 and 1996*a*, Chapter 9).

An alternative version of the median voter model is that where concern for the unemployed stems from a prudential, insurance motive, where the employed recognise that they themselves may be future recipients. The idea that voters determine their policy choices behind a genuine 'veil of ignorance', rather than a hypothetical veil as with the solidaristic welfare function, may provide an explanation as to why political support for unemployment compensation declined with the onset of recession. During the years of full employment of the 1950s and 1960s, support for the Welfare State persisted since, when unemployment was low, people remained uncertain whether they would be affected if we returned to unemployment of the level of the 1930s. By the time that the rise in unemployment in the 1980s had levelled off, however, people had a much better idea as to whether or not they were likely to be at risk and what was the probability of finding another job. The veil had been lifted. The majority found that they were not at risk, and they ceased to give as much weight to the risk of unemployment in their objective function. (This may of course have changed recently if there has been a rise in job insecurity.)

This is speculation. The main point to be made is that the explanation of trends in the income distribution cannot be complete without an analysis of public choice, and this cannot be treated simply as a routine application of a well-tried theory.

VII. CONCLUSIONS: BRINGING IT TOGETHER

My principal purpose here has been to argue that the economic analysis of the distribution of income is in need of further development before we can hope to give a definitive answer to the questions in which the ordinary person is interested – such as what determines the extent of inequality and why has inequality increased? This does not mean that current economic theory has nothing to contribute. It certainly offers insights into parts of the story, but what is required is for the different elements to be brought together. We need an overall framework, both conceptual and empirical, within which to fit the different mechanisms. The skill shift explanation for wage differentials is valuable, but it is only part of the story. The labour market cannot be seen as totally independent from the capital market. Both economic and political economy explanations have their place.

About 20 years ago, there was a spate of Presidential Addresses which were full of gloom about the state of economics. Since I began by criticising the profession for what I feel to have been its neglect of a central subject, I would like to end on a positive note. The first ground for optimism is the upsurge of interest in the recent past. The contributions which I have mentioned, and others not covered, are a welcome indication that income distribution is beginning to receive again the attention which it merits. The second is that there is evidence that economics is beginning to learn in this area from other disciplines. I have touched on social norms, where we can learn from the sociology of labour markets and from social psychology. I have discussed public choice, where we can learn from political science. A subject so central to social science as income distribution is unlikely to be one that we can solve on our own, and I take a receptiveness to outside ideas to be a sign of a discipline in good health.

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APPENDIX: SOURCES FOR FIGURES

Figure 2

United States

1947–67 for family (excluding unrelated individuals) gross income, unadjusted for family size, with family weights, from Nelson (1994), Table 2.1, linked at 1967 to 1967–92 for household gross income, unadjusted for household size, with household weights, from U.S. Department of Commerce (1993), Table B-3.

United Kingdom

1961–91 for equivalent household disposable income, with person weights, from Goodman and Webb (1994), page A2 (BHC); I am grateful to Alissa Goodman and Steven Webb for supplying comparable figures for 1992 and 1993.

Figure 3

United States and United Kingdom as above.

Japan

1980–91 supplied by Management and Coordination Agency, see Atkinson *et al.* (1995), Chapter 5.

Sweden

1975–91 for equivalent disposable income, with person weights, from Gustafsson and Palmer (1993), Annex.

Figure 4

France

1979, 1985 and 1989 for equivalent household (excluding households with retired head) disposable income, with person weights, Bourguignon and Martinez (1995).

Germany

1978, 1983, 1985, 1987 and 1990 for equivalent household (excluding households with non-German head) disposable income, with person weights, from Hauser and Becker (1993), Table 7, linked at 1983.

Italy

1977–91 for equivalent household disposable income, with household weights, from Brandolini and Sestito (1994), Table 2a; I am grateful to Andrea Brandolini for supplying a comparable figure for 1993.

Figure 5

Goodman and Webb (1994), pp. A2, A14 and A26.

Figure 6

Income as Fig. 2; earnings from Atkinson and Micklewright (1992), Table BE1. The earnings series covers all full-time workers.

Figure 7

New Earnings Survey 1979, Table 96, and 1990, Table 8 in Part A.